Lambda 1 continuation: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

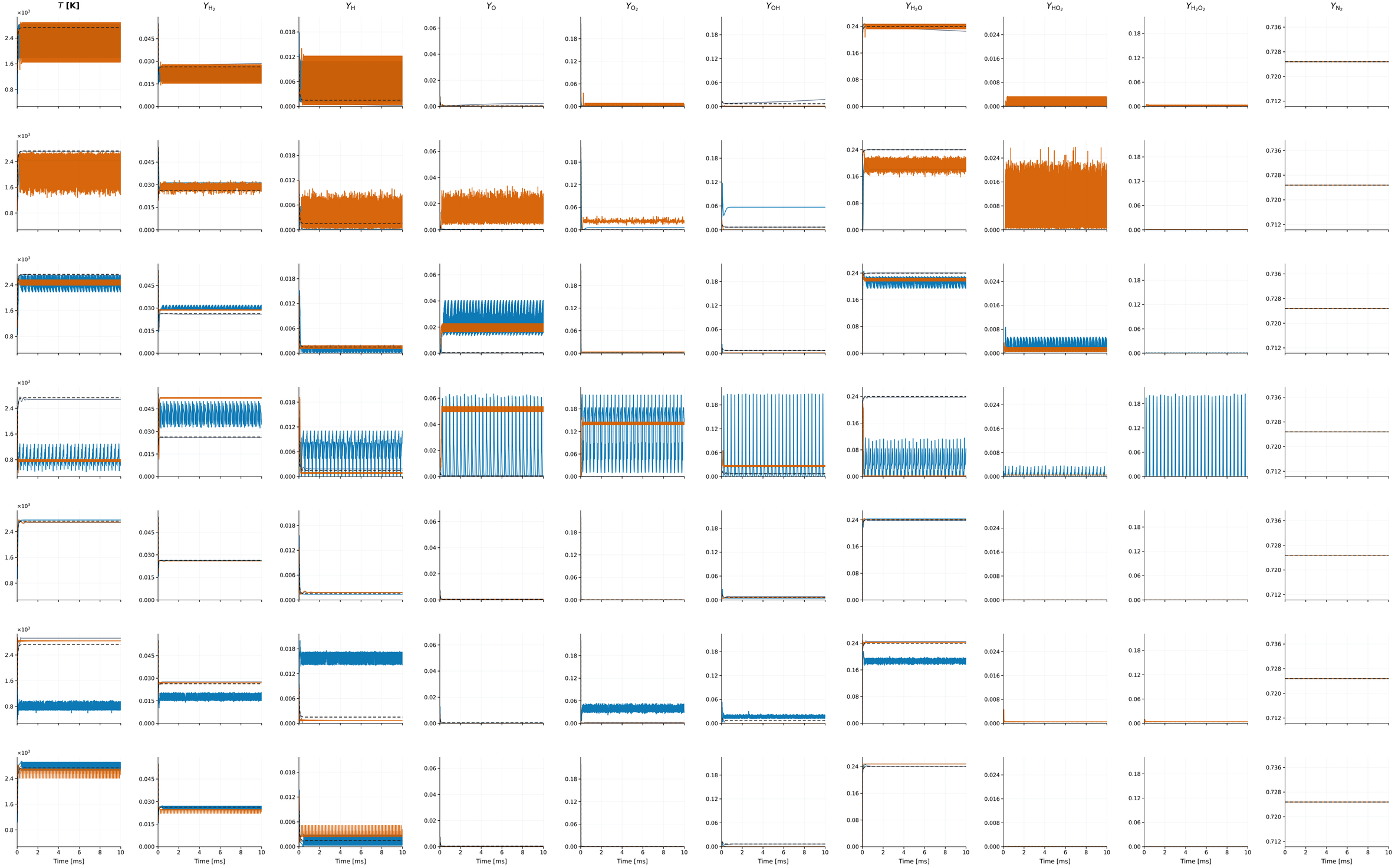
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 04 | Cantera vs NN NRMSE

T0=1317.5 K P0=1.244 atm phi0=1.984 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

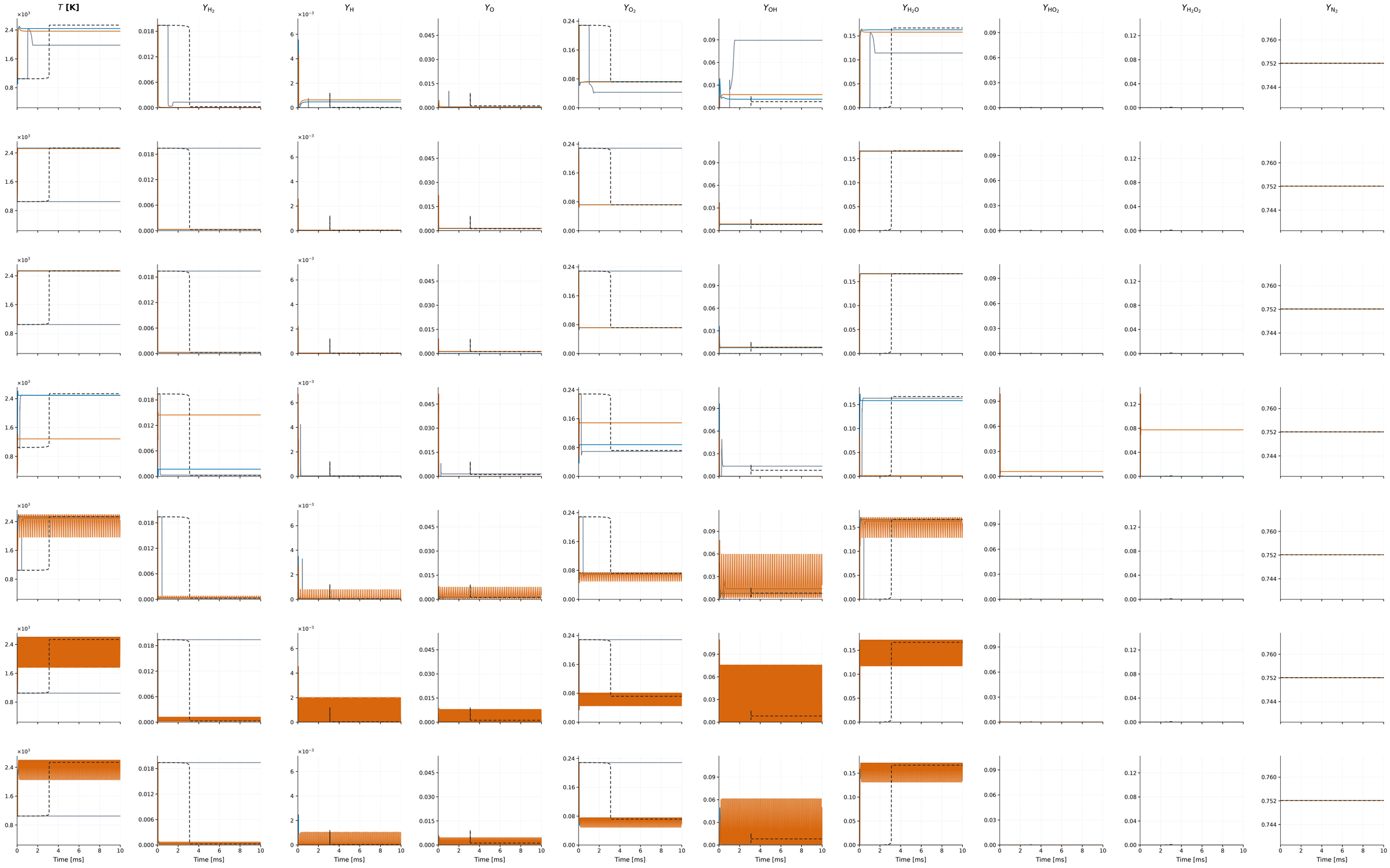
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
1 ANN10	soft10k	0.235	0.253	4.12	0.891	0.427	0.971	0.0296	28.3	551	0	Complete
1 ANN10	long	0.224	0.223	3.8	0.723	0.462	0.983	0.0228	451	1.38e+04	0	Complete
2 P-only DeepONet10	base10k	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	soft10k	0.109	0.2	0.888	0.462	1.24	7.23	0.253	0.999	5.42	0	Complete
2 P-only DeepONet10	long	0.266	0.11	2.1	30.9	2.23	0.979	0.182	2.31e+03	1.55e+03	0	Complete
3 ANN10 + P-only DeepONet10	base10k	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.12	0.151	0.679	45.4	0.614	0.988	0.118	933	3.44e+03	0	Complete
3 ANN10 + P-only DeepONet10	long	0.0847	0.0937	0.239	31.9	0.288	0.912	0.0749	311	396	0	Complete
4 T/P DeepONet10	base10k	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	soft10k	0.703	0.597	3.61	40.5	13.7	7.74	0.844	278	5.21e+05	0	Complete
4 T/P DeepONet10	long	0.714	0.982	0.763	89.9	13.6	2.91	0.988	161	0.758	0	Complete
5 ANN10 + T/P DeepONet10	base10k	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.0233	0.0207	0.323	0.481	0.182	0.356	0.0142	0.347	2	0	Complete
5 ANN10 + T/P DeepONet10	long	0.0133	0.0264	0.256	0.274	0.321	0.172	0.0161	0.664	0.282	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.716	0.372	9.46	0.718	3.56	1.49	0.213	0.563	0.733	0	Complete
6 T/P/ϕ DeepONet10	long	0.0453	0.035	0.544	0.996	0.389	1	0.0226	165	3.17e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.0447	0.0412	0.628	0.907	0.141	0.98	0.034	5.91	562	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.0444	0.0805	1.04	0.87	0.312	0.994	0.037	8.85	505	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 05 | Cantera vs NN NRMSE

T0=1049.0 K P0=4.666 atm phi0=0.674 | Repaired self-fed,10ms

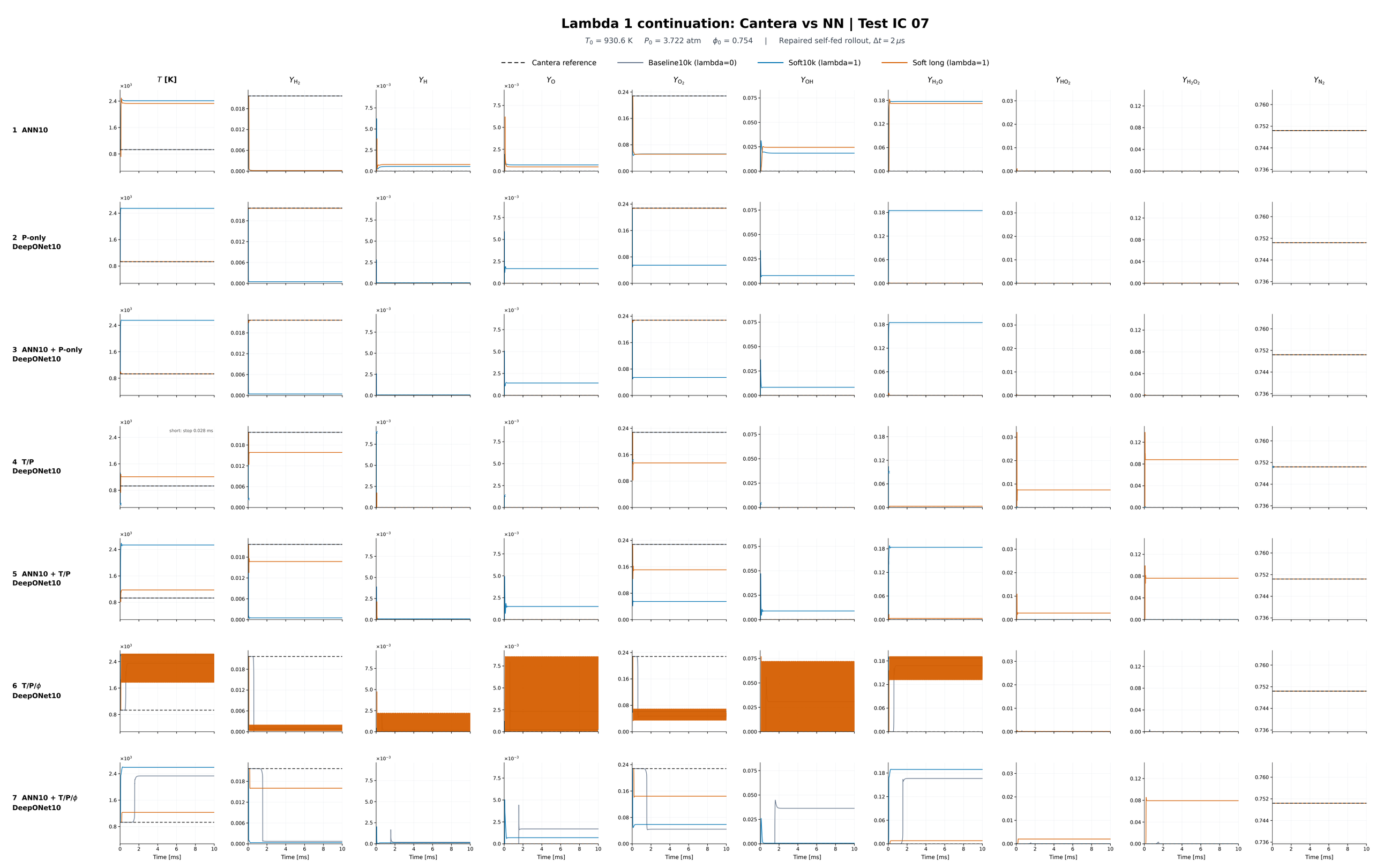
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
1 ANN10	**soft10k**	0.352	0.987	14.1	0.633	0.613	1.11	0.646	2.54	0.981	0	Complete
1 ANN10	**long**	0.338	0.983	18.7	0.707	0.614	1.81	0.629	0.976	0.994	0	Complete
2 P-only DeepONet10	**base10k**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	**soft10k**	0.375	0.981	1.42	0.759	0.617	0.677	0.664	0.948	0.998	0	Complete
2 P-only DeepONet10	**long**	0.371	0.977	1.71	0.925	0.616	0.738	0.66	1.28	1.07	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.376	0.982	1.51	0.685	0.617	0.665	0.666	0.903	0.997	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.373	0.979	1.65	0.778	0.617	0.701	0.663	1	1	0	Complete
4 T/P DeepONet10	**base10k**	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	**soft10k**	0.364	0.915	4.26	1.05	0.563	1.11	0.633	0.92	0.959	0	Complete
4 T/P DeepONet10	**long**	0.479	1.12	6.25	1.62	0.554	1.03	0.991	103	487	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.374	0.979	2.56	0.739	0.615	0.71	0.663	0.884	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.355	0.972	8.61	3.08	0.641	3.7	0.635	0.898	0.996	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.388	0.971	1	0.782	0.593	1	0.676	0.995	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.352	0.961	20.4	4.07	0.643	5.15	0.624	1.64	0.993	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.382	0.978	2.2	0.683	0.603	0.962	0.674	0.89	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.359	0.971	10.2	2.11	0.633	3.78	0.641	1.19	0.994	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 07 | Cantera vs NN NRMSE

T0=930.6 K P0=3.722 atm phi0=0.754 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

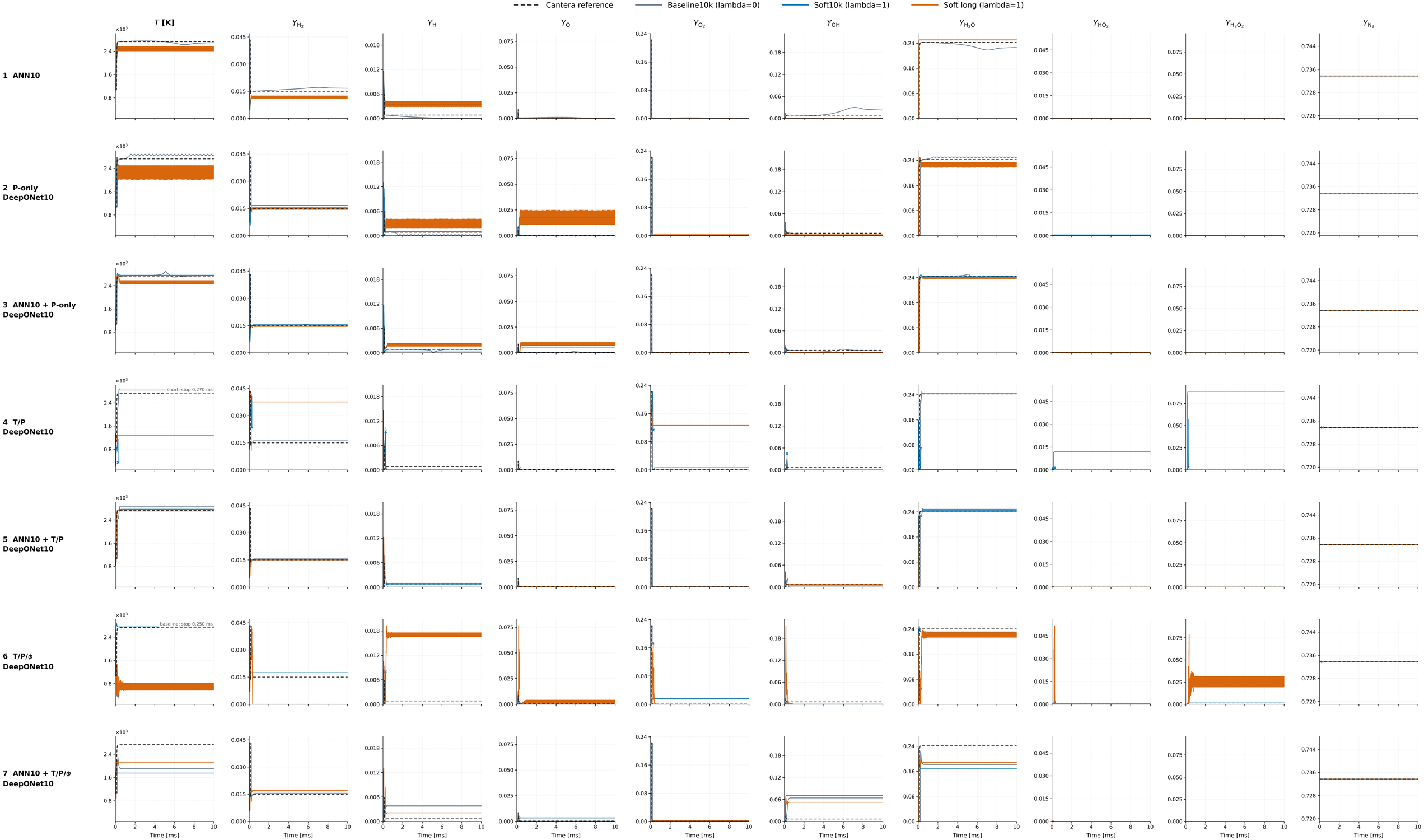
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
1 ANN10	soft10k	1.58	0.99	1.60e+08	9.77e+07	0.769	4.13e+09	1.06e+06	199	18.3	0	Complete
1 ANN10	long	1.49	0.988	2.23e+08	7.29e+07	0.77	5.37e+09	1.03e+06	137	15.5	0	Complete
2 P-only DeepONet10	base10k	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	soft10k	1.73	0.977	2.26e+07	2.25e+08	0.758	1.77e+09	1.11e+06	52.6	7.03	0	Complete
2 P-only DeepONet10	long	4.81e-03	2.81e-03	3.15e+03	441	2.81e-03	4.27e+03	2.29e+03	65.5	738	0	Complete
3 ANN10 + P-only DeepONet10	base10k	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	1.74	0.979	2.24e+07	1.91e+08	0.76	1.88e+09	1.11e+06	54.5	5.62	0	Complete
3 ANN10 + P-only DeepONet10	long	3.84e-03	2.08e-03	9.22e+03	846	1.83e-03	8.23e+03	2.09e+03	44.5	284	0	Complete
4 T/P DeepONet10	base10k	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
4 T/P DeepONet10	long	0.299	0.267	1.43e+07	550	0.406	1.81e+04	1.92e+04	1.55e+04	2.22e+05	0	Complete
5 ANN10 + T/P DeepONet10	base10k	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	1.72	0.975	3.66e+07	2.04e+08	0.757	2.00e+09	1.10e+06	37.9	2.61	0	Complete
5 ANN10 + T/P DeepONet10	long	0.262	0.227	1.91e+07	2.85e+04	0.336	3.71e+05	1.83e+04	5.76e+03	1.92e+05	0	Complete
6 T/P/ϕ DeepONet10	base10k	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	soft10k	1.79	0.966	2.92e+06	3.23e+07	0.73	5.19e+05	1.12e+06	11.5	1.01	0	Complete
6 T/P/ϕ DeepONet10	long	1.49	0.951	2.37e+08	6.69e+08	0.767	7.97e+09	1.01e+06	129	10.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	1.78	0.98	3.71e+07	1.04e+08	0.74	5.22e+08	1.13e+06	30.4	1.59	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.319	0.258	2.90e+04	691	0.363	2.71e+03	4.69e+04	4.2e+03	1.99e+05	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 18 | Cantera vs NN NRMSE

T0=1069.1 K P0=2.170 atm phi0=1.544 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

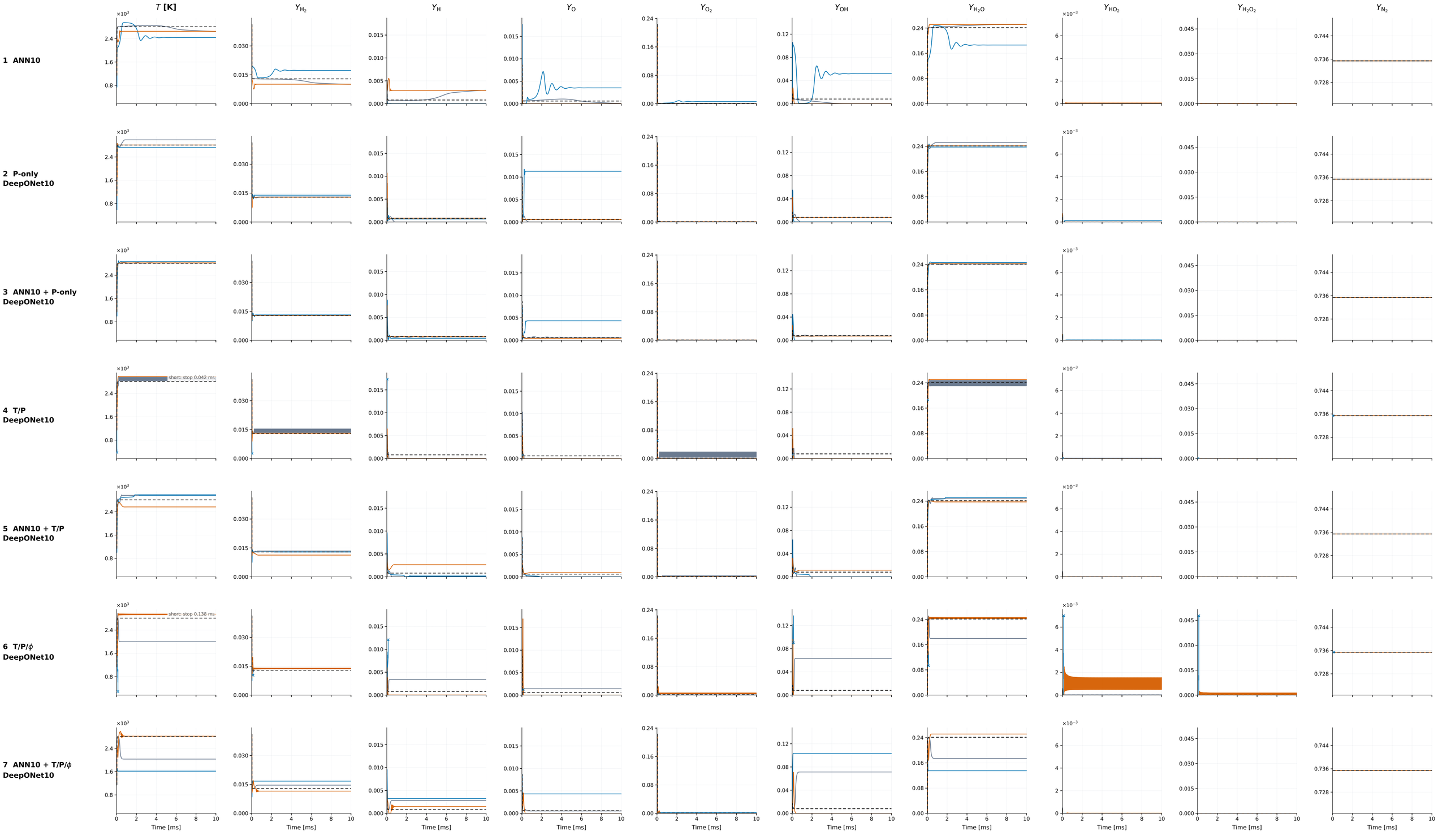
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
1 ANN10	**soft10k**	0.0856	0.274	3.21	0.99	0.697	1	0.0878	2.71	73.7	0	Complete
1 ANN10	**long**	0.0938	0.265	3.33	0.978	0.645	0.999	0.0822	6.44	144	0	Complete
2 P-only DeepONet10	**base10k**	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	**soft10k**	0.0869	0.199	0.927	34.1	0.631	1.02	0.0847	34.5	2.07	0	Complete
2 P-only DeepONet10	**long**	0.206	0.159	2.9	37.4	0.711	0.735	0.11	7.42	3.2	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.0228	0.178	0.954	8.55	0.672	0.981	0.0743	13.8	1.83	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0796	0.161	1.31	14.6	0.703	0.865	0.0796	7.19	1.55	0	Complete
4 T/P DeepONet10	**base10k**	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.270 ms
4 T/P DeepONet10	**long**	0.528	1.43	1	1	5.02	1	0.993	832	7.66e+04	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.0329	0.184	0.893	0.514	0.761	0.408	0.0793	1.11	0.974	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0213	0.14	0.662	0.471	0.561	0.252	0.0599	1.34	1.29	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.250 ms
6 T/P/ϕ DeepONet10	**soft10k**	0.0553	0.226	1.04	1.01	0.995	1	0.102	23	1.28e+03	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.749	0.963	19.1	10.9	0.819	2.33	0.153	125	2.30e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.358	0.195	3.78	5.81	0.784	9.69	0.311	1.18	1.15	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.221	0.198	1.63	6.22	0.621	6.84	0.232	1.36	1.22	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 32 | Cantera vs NN NRMSE

T0=1140.7 K P0=3.354 atm phi0=1.463 | Repaired self-fed,10ms

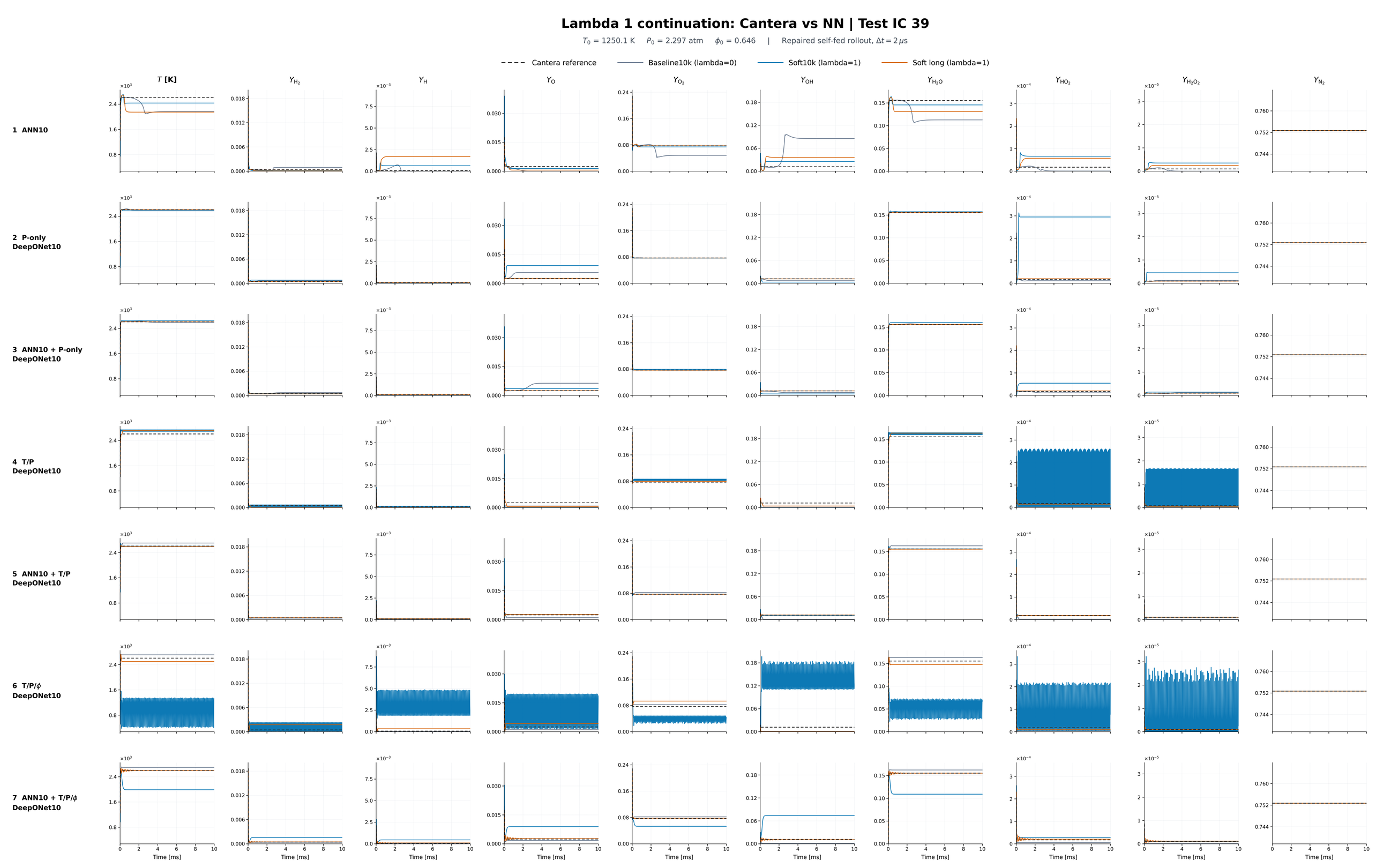
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
1 ANN10	**soft10k**	0.126	0.312	0.991	4.6	0.374	5.21	0.216	0.765	0.904	0	Complete
1 ANN10	**long**	0.0604	0.216	2.65	0.963	0.214	1	0.0438	5.96	182	0	Complete
2 P-only DeepONet10	**base10k**	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	**soft10k**	0.0366	0.0924	0.352	16.8	0.392	0.997	0.0296	10.6	2.02	0	Complete
2 P-only DeepONet10	**long**	0.0112	0.0207	0.275	0.212	0.082	0.105	6.26e-03	0.564	0.606	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.0308	0.0574	0.59	5.83	0.385	1.03	0.033	3.59	3.04	0	Complete
3 ANN10 + P-only DeepONet10	**long**	6.46e-03	0.0408	0.172	0.362	0.246	0.123	0.0152	0.725	0.852	0	Complete
4 T/P DeepONet10	**base10k**	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.042 ms
4 T/P DeepONet10	**long**	0.0625	0.0502	1.01	1.03	0.32	1.02	0.0441	0.806	0.959	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.06	0.0683	0.971	0.981	0.395	1.01	0.0468	0.973	3.89	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0853	0.123	2.2	0.542	0.11	0.45	0.0177	0.424	0.602	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.138 ms
6 T/P/ϕ DeepONet10	**long**	0.0529	0.0863	1	1.67	0.415	1.16	0.0359	92.2	942	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.424	0.3	2.99	5.91	0.484	11.7	0.44	1.73	2.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.0243	0.114	0.854	1.07	0.317	1.15	0.0523	1.95	11.2	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 39 | Cantera vs NN NRMSE

T0=1250.1 K P0=2.297 atm phi0=0.646 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
1 ANN10	**soft10k**	0.0665	0.517	5.92	0.548	0.0509	1.15	0.0653	2.65	2.51	0	Complete
1 ANN10	**long**	0.172	0.426	17	0.728	0.015	2.03	0.151	2.11	1.49	0	Complete
2 P-only DeepONet10	**base10k**	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	**soft10k**	0.0247	0.904	0.643	2.66	0.0687	0.732	0.0409	15	3.55	0	Complete
2 P-only DeepONet10	**long**	3.98e-03	0.179	0.21	0.0944	0.0137	0.0231	8.06e-03	0.192	0.141	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.0237	0.497	0.701	0.516	0.0415	0.668	0.0389	1.98	0.469	0	Complete
3 ANN10 + P-only DeepONet10	**long**	1.68e-03	0.0891	0.312	0.0632	6.56e-03	0.0233	3.10e-03	0.152	0.0755	0	Complete
4 T/P DeepONet10	**base10k**	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	**soft10k**	0.0408	0.201	0.703	0.966	0.0937	0.998	0.0467	4.83	7.85	0	Complete
4 T/P DeepONet10	**long**	0.0432	0.464	0.831	0.807	0.034	0.662	0.053	0.313	0.634	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	6.23e-03	0.116	0.353	0.176	9.54e-03	0.0848	7.54e-03	0.113	0.07	0	Complete
5 ANN10 + T/P DeepONet10	**long**	5.99e-03	0.125	0.333	0.106	0.0118	0.0384	7.52e-03	0.0895	0.0741	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.624	1.27	32.7	4.23	0.499	10.7	0.638	4.51	8.29	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.0414	1.44	2.93	0.687	0.206	0.986	0.0498	0.469	0.101	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.234	1.25	4.06	2.53	0.303	5.21	0.294	0.603	0.124	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	5.93e-03	0.0991	0.43	0.131	0.0139	0.11	6.49e-03	0.208	0.219	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference | Baseline10k (lambda=0) | Soft10k (lambda=1) | Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

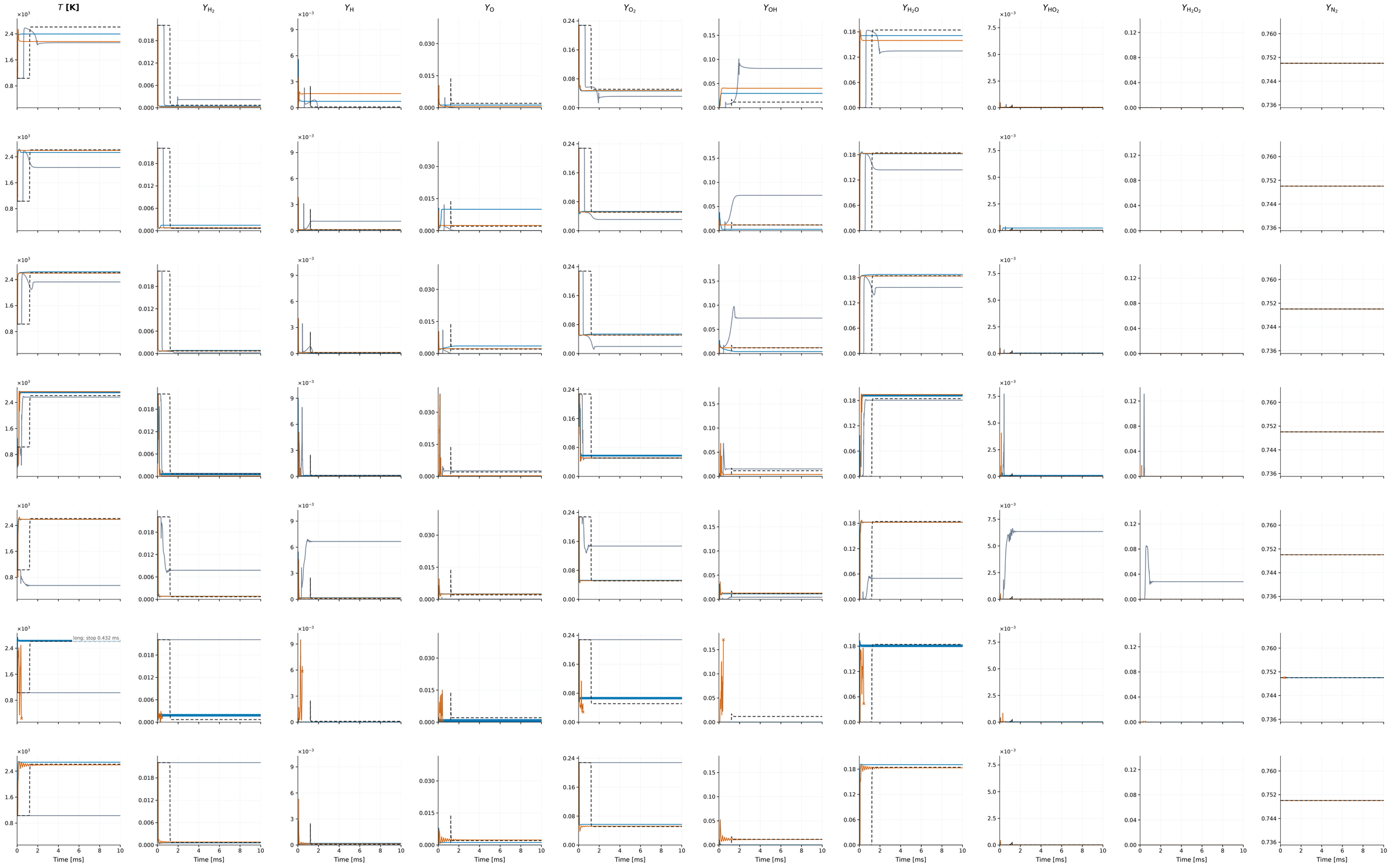
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 44 | Cantera vs NN NRMSE

T0=1031.1 K P0=2.272 atm phi0=0.770 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
1 ANN10	**soft10k**	0.203	0.953	6.1	0.491	0.663	1.76	0.344	1.47	0.99	0	Complete
1 ANN10	**long**	0.231	0.959	14	0.77	0.654	2.69	0.343	1.47	0.984	0	Complete
2 P-only DeepONet10	**base10k**	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	**soft10k**	0.21	0.92	1.05	4.02	0.643	0.804	0.359	8.17	1.03	0	Complete
2 P-only DeepONet10	**long**	0.215	0.942	1	0.494	0.649	0.373	0.361	0.95	1.02	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.218	0.942	1.19	0.847	0.646	0.715	0.365	1.53	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.215	0.944	1.01	0.458	0.65	0.381	0.362	0.917	0.988	0	Complete
4 T/P DeepONet10	**base10k**	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	**soft10k**	0.221	0.902	3.95	1.37	0.596	1	0.359	2.18	1.9	0	Complete
4 T/P DeepONet10	**long**	0.225	0.924	2.52	1.34	0.623	0.768	0.364	4.45	50.5	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.214	0.943	1.58	0.53	0.648	0.383	0.361	0.844	0.978	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.209	0.925	1.62	0.518	0.639	0.44	0.353	1.07	0.988	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.222	0.912	0.975	0.782	0.618	1	0.361	0.959	0.999	0	Complete
6 T/P/ϕ DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.432 ms
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.227	0.951	1.28	0.518	0.632	0.982	0.377	0.841	0.992	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.209	0.927	1.48	0.517	0.641	0.496	0.353	1.01	0.996	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

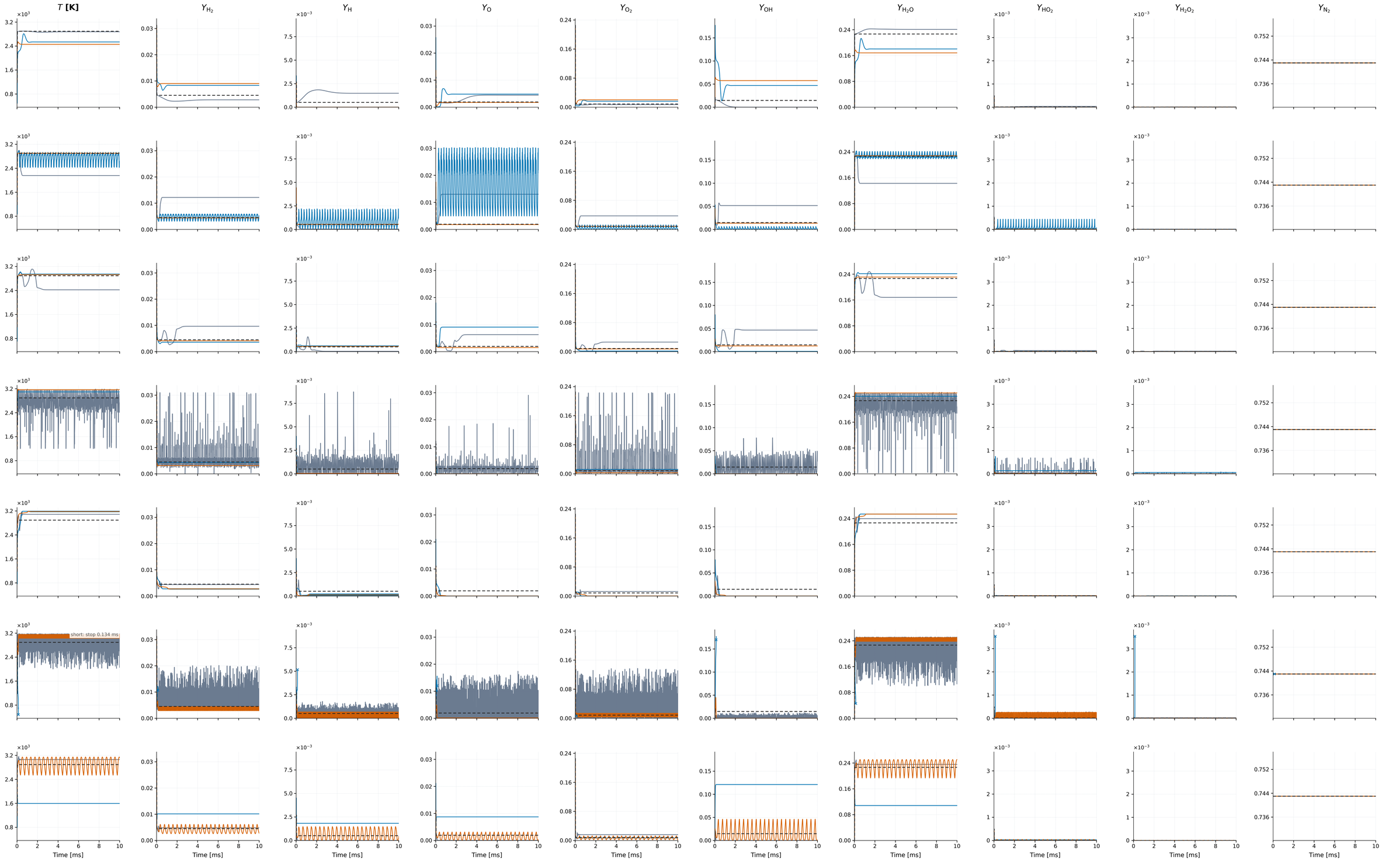
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 45 | Cantera vs NN NRMSE

T0=1181.9 K P0=4.842 atm phi0=1.099 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
1 ANN10	soft10k	0.128	0.799	1.01	1.56	0.664	2.46	0.211	1.01	0.647	0	Complete
1 ANN10	long	0.15	0.953	1	0.214	0.953	2.93	0.258	1.06	0.684	0	Complete
2 P-only DeepONet10	base10k	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	soft10k	0.0717	0.228	1.31	8.8	0.519	0.872	0.0335	9.57	1.59	0	Complete
2 P-only DeepONet10	long	0.0151	0.159	0.159	0.174	0.445	0.0988	0.0271	0.84	0.668	0	Complete
3 ANN10 + P-only DeepONet10	base10k	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.0203	0.215	0.247	3.61	0.59	0.962	0.0664	0.914	1	0	Complete
3 ANN10 + P-only DeepONet10	long	0.0181	0.172	0.135	0.249	0.449	0.166	0.0316	0.815	0.717	0	Complete
4 T/P DeepONet10	base10k	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	soft10k	0.07	0.17	1.01	0.997	0.525	0.998	0.065	9.21	41.6	0	Complete
4 T/P DeepONet10	long	0.0959	0.325	0.904	0.998	0.6	0.999	0.11	1.06	0.864	0	Complete
5 ANN10 + T/P DeepONet10	base10k	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.1	0.393	0.791	1.02	0.723	1.04	0.121	0.91	2.62	0	Complete
5 ANN10 + T/P DeepONet10	long	0.0985	0.393	0.776	0.988	0.744	0.983	0.119	0.882	0.866	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
6 T/P/ϕ DeepONet10	long	0.0816	0.303	0.75	0.968	0.545	1	0.0964	5.73	1.02	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.448	1.19	2.59	3.51	0.432	7.24	0.522	1.67	1.27	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.0743	0.306	1.06	0.611	0.531	1.08	0.0896	1.45	2.11	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

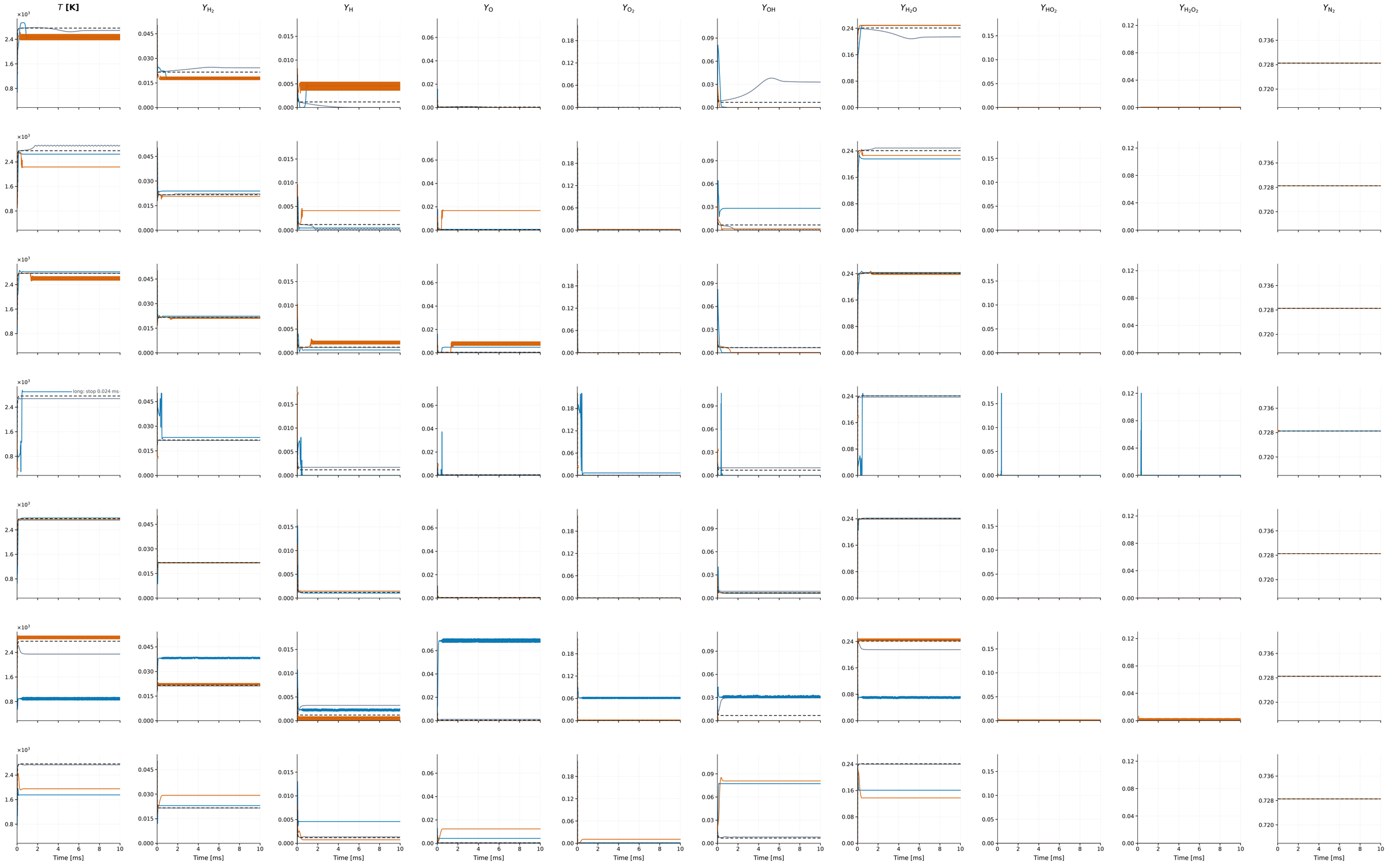
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 53 | Cantera vs NN NRMSE

T0=1257.8 K P0=2.103 atm phi0=1.799 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

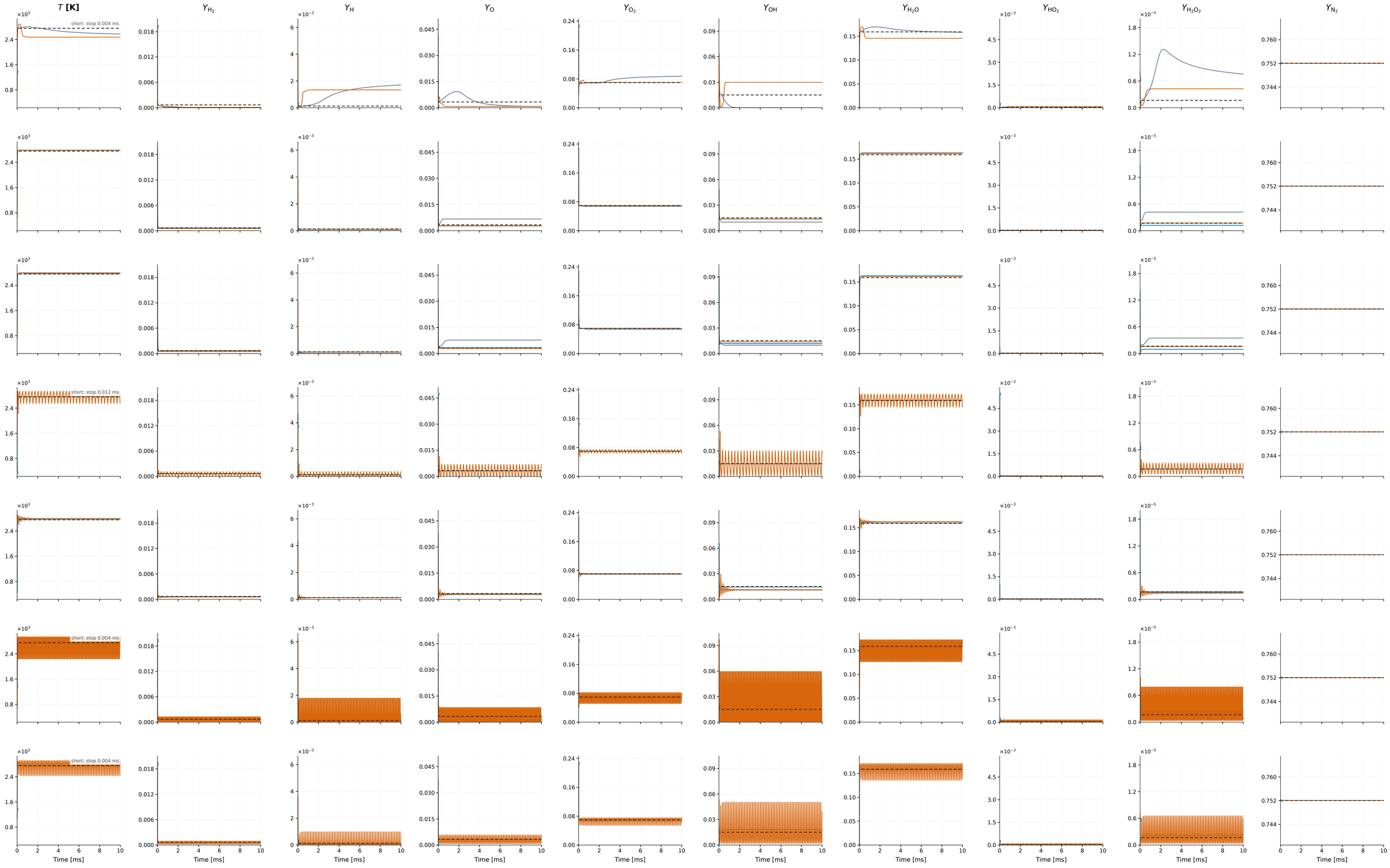
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
1 ANN10	soft10k	0.108	0.174	2.66	1.07	0.52	1.55	0.0561	6.74	665	0	Complete
1 ANN10	long	0.108	0.177	2.78	0.838	0.157	0.994	0.0335	20.9	1.15e+03	0	Complete
2 P-only DeepONet10	base10k	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	soft10k	0.0545	0.126	0.661	1.43	1.14	3.18	0.118	0.997	3.15	0	Complete
2 P-only DeepONet10	long	0.188	0.0542	2.38	32.2	0.519	0.74	0.0638	12.7	0.911	0	Complete
3 ANN10 + P-only DeepONet10	base10k	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.0354	0.0549	0.522	8.6	0.598	1.2	0.0423	36.5	4.79	0	Complete
3 ANN10 + P-only DeepONet10	long	0.0698	0.0264	0.961	15.4	0.171	0.88	0.0142	28.8	6.01	0	Complete
4 T/P DeepONet10	base10k	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	soft10k	0.146	0.208	1.21	2.27	4.02	1.13	0.184	1.1e+03	2.66e+04	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
5 ANN10 + T/P DeepONet10	base10k	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.0383	0.0367	0.688	0.32	0.281	0.26	0.0152	0.549	0.884	0	Complete
5 ANN10 + T/P DeepONet10	long	0.015	0.0165	0.283	0.12	0.12	0.0974	5.57e-03	0.348	0.279	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.678	0.766	0.948	137	6.45	3.55	0.71	16.5	0.842	0	Complete
6 T/P/ϕ DeepONet10	long	0.0505	0.0405	0.767	0.931	0.341	0.995	0.025	200	7.82e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.366	0.0672	2.85	7.61	0.209	10.3	0.332	1.09	1.91	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.292	0.347	0.396	23.7	1.09	10.8	0.425	0.656	0.397	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1) · Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 57 | Cantera vs NN NRMSE

T0=1412.0 K P0=4.083 atm phi0=0.677 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
1 ANN10	long	0.1	0.685	9.03	0.779	0.0264	0.977	0.0841	1.73	1.51	0	Complete
2 P-only DeepONet10	base10k	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	soft10k	0.0213	0.623	0.279	0.214	0.0454	0.11	0.0337	0.279	0.303	0	Complete
2 P-only DeepONet10	long	0.0148	0.317	0.342	0.203	0.0237	0.0812	0.0198	0.16	0.0772	0	Complete
3 ANN10 + P-only DeepONet10	base10k	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.0252	0.645	0.208	0.181	0.0389	0.274	0.0414	0.201	0.443	0	Complete
3 ANN10 + P-only DeepONet10	long	0.0142	0.276	0.31	0.188	0.0134	0.0913	0.0189	0.0872	0.0662	0	Complete
4 T/P DeepONet10	base10k	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
4 T/P DeepONet10	long	0.0446	0.4	0.938	0.647	0.0431	0.593	0.0541	0.451	0.453	0	Complete
5 ANN10 + T/P DeepONet10	base10k	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.024	0.426	0.429	0.289	0.0326	0.28	0.0326	0.696	0.18	0	Complete
5 ANN10 + T/P DeepONet10	long	0.0189	0.192	0.329	0.255	0.0241	0.299	0.0249	0.16	0.213	0	Complete
6 T/P/ϕ DeepONet10	base10k	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	long	0.102	0.537	4.62	0.977	0.141	1.52	0.116	1.98	1.84	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
7 ANN10 + T/P/ϕ DeepONet10	long	0.0657	0.387	2.41	0.599	0.106	1.16	0.0813	1.39	1.48	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

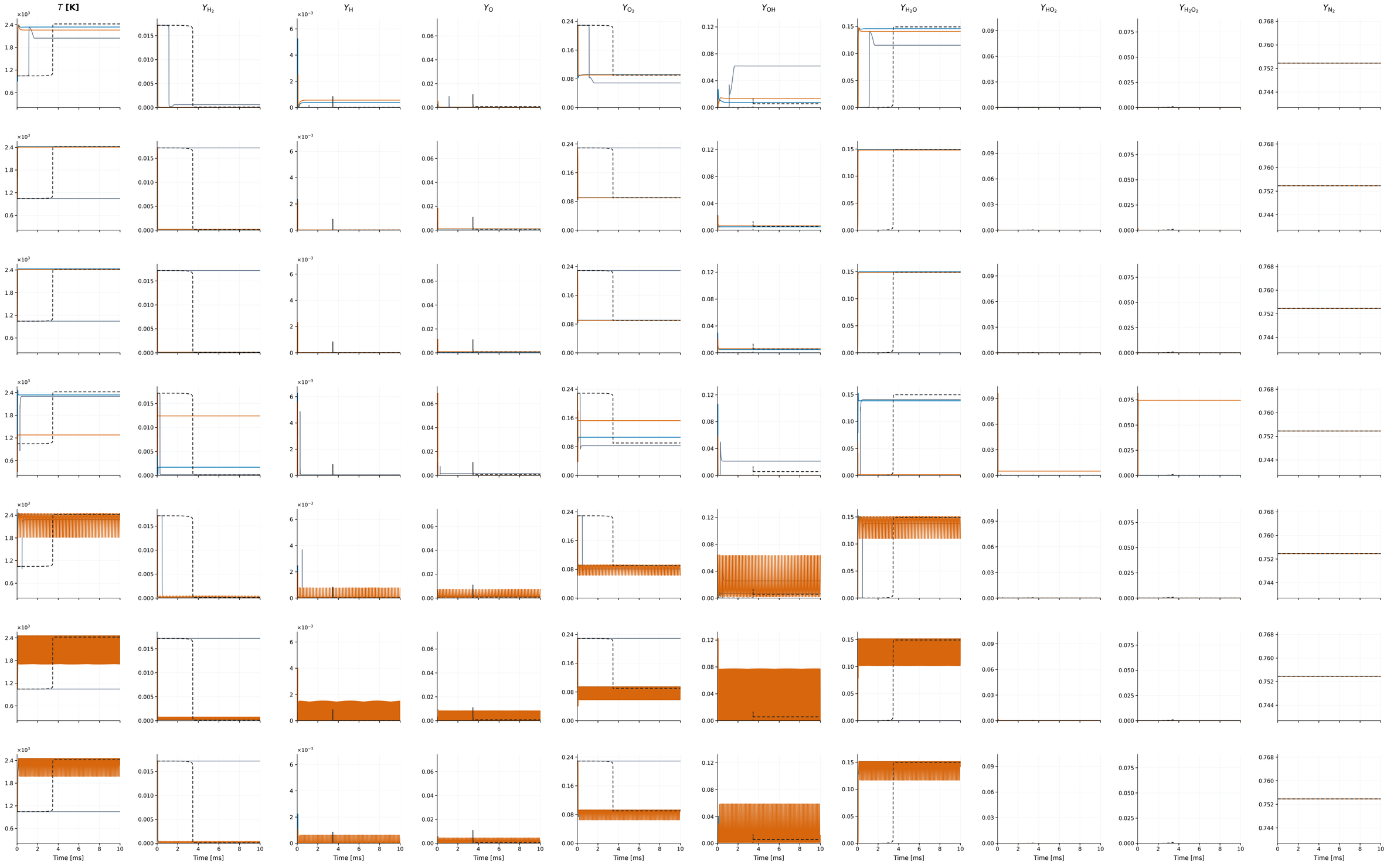
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 64 | Cantera vs NN NRMSE

T0=1045.3 K P0=4.301 atm phi0=0.595 | Repaired self-fed,10ms

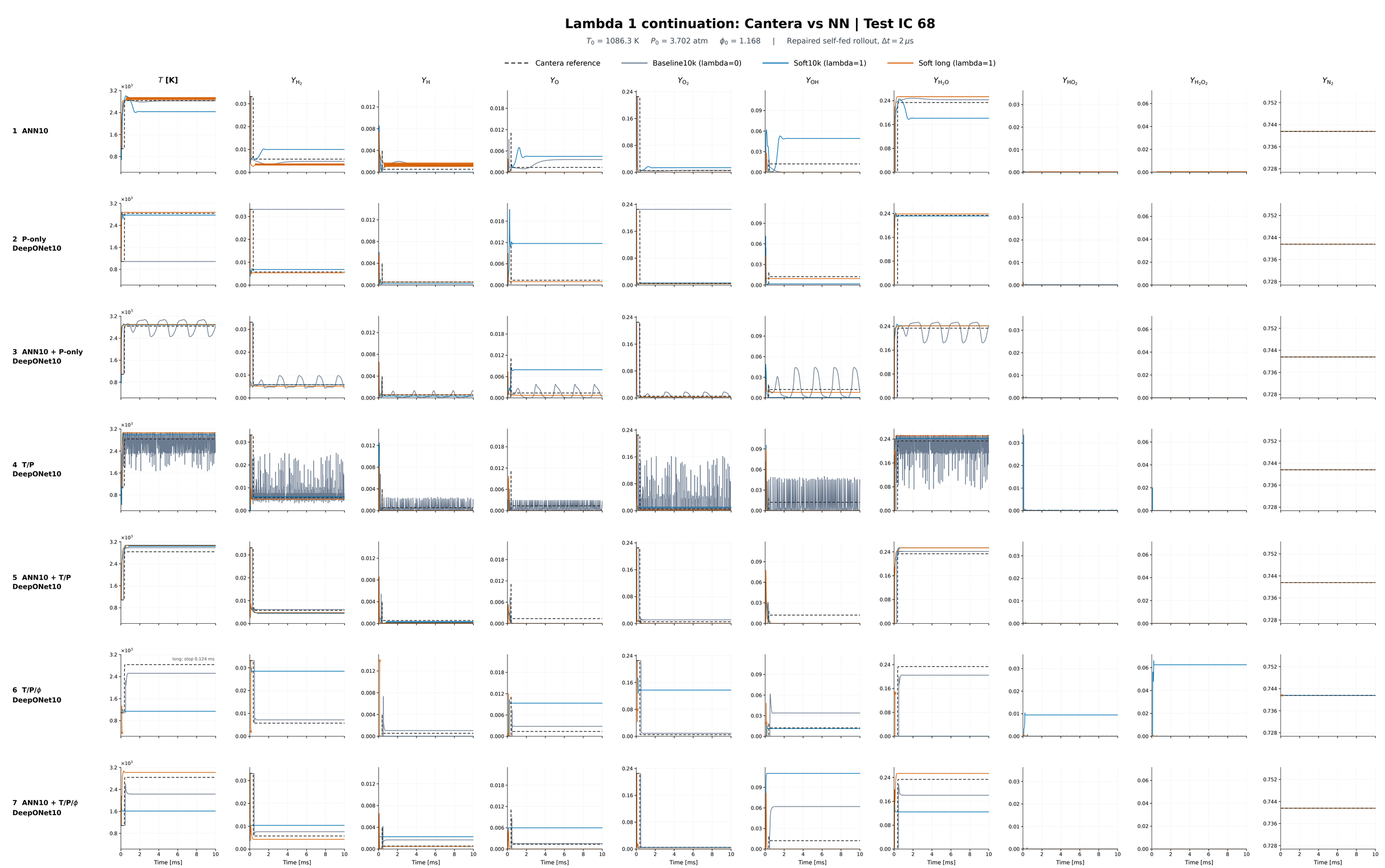
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
1 ANN10	**soft10k**	0.367	0.991	20.5	0.681	0.523	1.12	0.702	3.92	0.977	0	Complete
1 ANN10	**long**	0.35	0.986	30	0.747	0.524	2.19	0.68	1	0.995	0	Complete
2 P-only DeepONet10	**base10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	**soft10k**	0.391	0.987	2.16	0.93	0.527	0.659	0.722	1.02	0.999	0	Complete
2 P-only DeepONet10	**long**	0.385	0.981	2.17	1.18	0.527	0.838	0.714	1.23	1.08	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.393	0.988	2.18	0.703	0.527	0.666	0.724	0.916	0.996	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.388	0.984	2.22	0.956	0.527	0.77	0.718	1.08	1.01	0	Complete
4 T/P DeepONet10	**base10k**	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	**soft10k**	0.368	0.906	8.67	1.14	0.477	1.28	0.671	0.892	0.954	0	Complete
4 T/P DeepONet10	**long**	0.455	1.03	8.68	1.98	0.439	1.05	0.99	91.3	498	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.387	0.985	3.42	0.899	0.529	0.933	0.717	0.898	0.998	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.365	0.979	13.1	4.57	0.56	5.79	0.682	0.912	0.996	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.399	0.978	1.03	0.793	0.512	1	0.73	0.987	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.358	0.968	26.8	6.58	0.567	7.89	0.668	1.94	0.991	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.395	0.983	2.89	0.768	0.519	0.909	0.726	0.894	0.998	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.368	0.976	11.3	3.18	0.549	5.32	0.688	1.2	0.993	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 68 | Cantera vs NN NRMSE

T0=1086.3 K P0=3.702 atm phi0=1.168 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

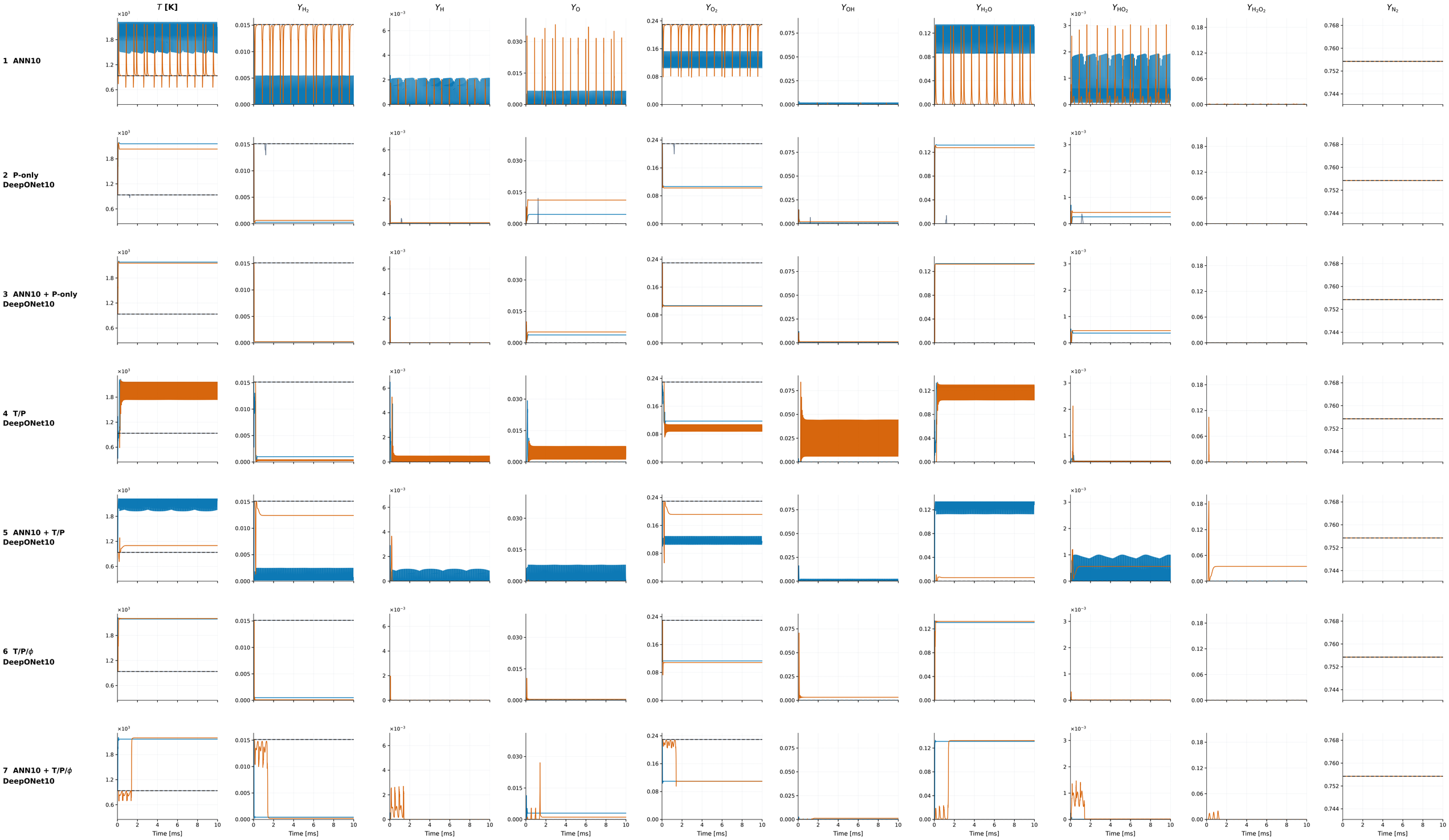
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
1 ANN10	soft10k	0.16	0.728	1.3	2.26	0.911	2.98	0.265	1.02	0.879	0	Complete
1 ANN10	long	0.101	0.683	1.86	1	0.908	1.03	0.203	4.89	27.4	0	Complete
2 P-only DeepONet10	base10k	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	soft10k	0.11	0.59	0.657	7.58	0.918	0.889	0.183	4.68	1.04	0	Complete
2 P-only DeepONet10	long	0.115	0.591	0.36	0.38	0.913	0.285	0.188	1.54	1.31	0	Complete
3 ANN10 + P-only DeepONet10	base10k	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.112	0.58	0.598	4.73	0.908	1.01	0.186	2.3	1.04	0	Complete
3 ANN10 + P-only DeepONet10	long	0.116	0.595	0.357	0.525	0.916	0.38	0.189	1.19	1.02	0	Complete
4 T/P DeepONet10	base10k	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	soft10k	0.118	0.584	1.62	1.04	0.89	1.1	0.178	23.1	37.5	0	Complete
4 T/P DeepONet10	long	0.135	0.558	1.18	1.06	0.858	1.08	0.191	1.08	1.02	0	Complete
5 ANN10 + T/P DeepONet10	base10k	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.129	0.587	0.974	1.06	0.913	1.09	0.193	1.01	1.33	0	Complete
5 ANN10 + T/P DeepONet10	long	0.135	0.574	0.999	1.04	0.892	1.08	0.195	1.07	0.997	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.597	2.62	1	5.8	2.98	0.161	1	395	7.32e+03	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.432	0.728	3.24	3.41	0.909	8.19	0.478	1.22	0.993	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.128	0.586	0.444	1.04	0.891	1.1	0.197	1.29	0.998	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 69 | Cantera vs NN NRMSE

T0=935.0 K P0=2.045 atm phi0=0.523 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

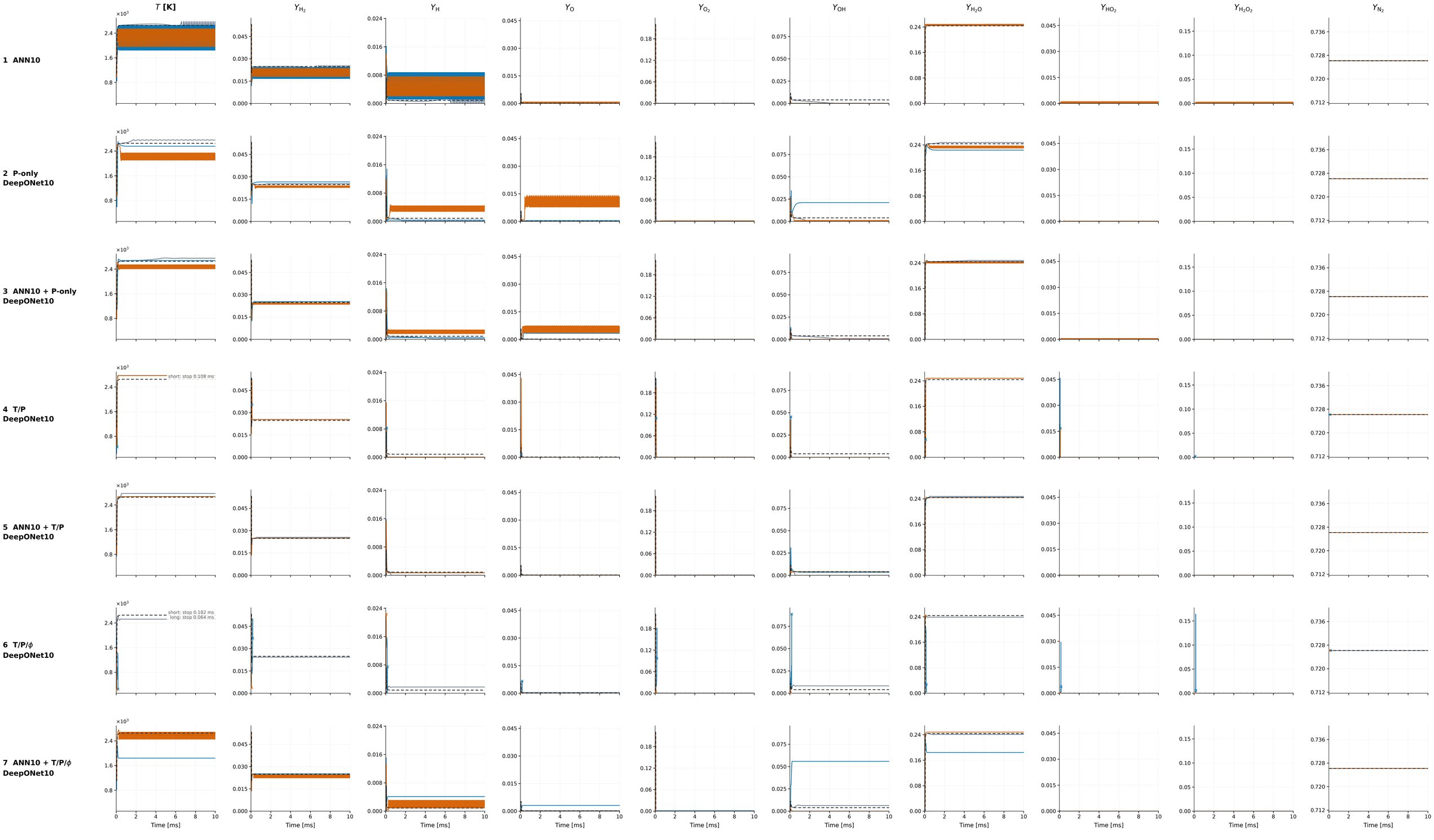
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
1 ANN10	**soft10k**	1.17	0.89	1.79e+08	2.82e+08	0.468	2.36e+08	1.76e+06	2.52e+03	2.82e+03	0	Complete
1 ANN10	**long**	0.286	0.23	1.18e+08	4.06e+08	0.13	2.65e+07	4.51e+05	1.68e+03	6.38e+03	0	Complete
2 P-only DeepONet10	**base10k**	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	**soft10k**	1.31	0.98	1.37e+07	4.63e+08	0.534	3.09e+08	1.96e+06	1.15e+03	89.1	0	Complete
2 P-only DeepONet10	**long**	1.17	0.957	2.99e+07	1.18e+09	0.552	5.86e+08	1.90e+06	1.88e+03	58.3	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	1.33	0.985	1.61e+07	3.94e+08	0.533	1.74e+08	1.98e+06	1.63e+03	114	0	Complete
3 ANN10 + P-only DeepONet10	**long**	1.3	0.983	1.23e+07	5.43e+08	0.54	3.67e+08	1.97e+06	2.04e+03	111	0	Complete
4 T/P DeepONet10	**base10k**	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	**soft10k**	1.29	0.927	9.81e+07	2.14e+08	0.485	6.39e+08	1.87e+06	188	808	0	Complete
4 T/P DeepONet10	**long**	1.15	0.968	1.13e+08	4.38e+08	0.557	6.62e+09	1.78e+06	354	4.30e+04	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	1.28	0.956	8.49e+07	3.04e+08	0.506	3.21e+08	1.91e+06	1.46e+03	720	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.172	0.186	8.34e+07	3.10e+06	0.169	1.73e+06	8.62e+04	2.46e+03	4.03e+05	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	1.35	0.966	8.28e+05	3.95e+07	0.507	2.01e+06	1.95e+06	28.3	0.911	0	Complete
6 T/P/ϕ DeepONet10	**long**	1.35	0.99	1.85e+07	7.21e+07	0.527	1.28e+09	1.97e+06	69.2	11.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	1.32	0.97	4.01e+06	3.20e+08	0.521	6.52e+07	1.95e+06	49	6.99	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	1.25	0.916	1.40e+08	1.60e+08	0.485	3.20e+08	1.83e+06	1.36e+03	3.08e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 73 | Cantera vs NN NRMSE

T0=1106.0 K P0=1.911 atm phi0=1.915 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0381	0.0487	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
1 ANN10	soft10k	0.204	0.223	5.82	0.976	0.319	0.99	0.0311	9.53	221	0	Complete
1 ANN10	long	0.11	0.131	3.29	1.1	0.218	0.995	0.0241	48.3	2.73e+03	0	Complete
2 P-only DeepONet10	base10k	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	soft10k	0.0523	0.0791	1	1.2	0.344	4.14	0.0852	1.55	4.32	0	Complete
2 P-only DeepONet10	long	0.176	0.0803	3.28	46	0.413	0.8	0.0542	11	3.03	0	Complete
3 ANN10 + P-only DeepONet10	base10k	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.0288	0.0345	0.833	12.7	0.0845	0.954	6.57e-03	16.5	6.4	0	Complete
3 ANN10 + P-only DeepONet10	long	0.0716	0.057	1.49	22.2	0.354	0.964	0.0305	34.6	33.5	0	Complete
4 T/P DeepONet10	base10k	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
4 T/P DeepONet10	long	0.0565	0.0828	1.02	6.94	0.648	1.05	0.0641	46.7	2.88	0	Complete
5 ANN10 + T/P DeepONet10	base10k	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.0159	0.0735	0.765	0.663	0.541	0.331	0.0438	1.13	0.993	0	Complete
5 ANN10 + T/P DeepONet10	long	0.011	0.0508	0.558	0.635	0.368	0.092	0.0297	0.932	1.11	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.182 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.064 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.305	0.0749	3.57	12.3	0.57	12.8	0.247	1.11	1.13	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.0361	0.0801	1.3	0.972	0.48	0.989	0.0435	2.06	74.1	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference | Baseline10k (lambda=0) | Soft10k (lambda=1) | Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

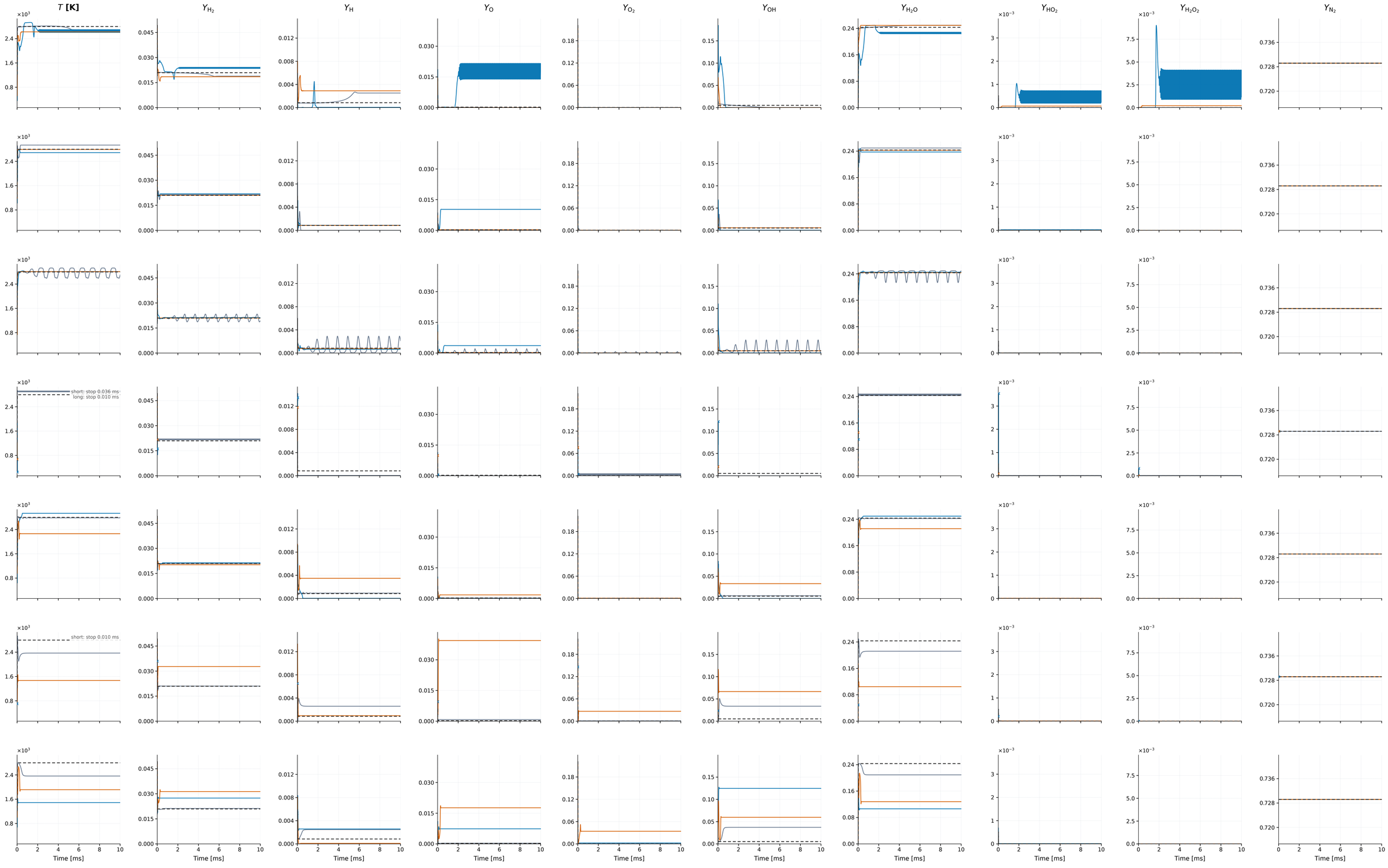
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 74 | Cantera vs NN NRMSE

T0=1222.9 K P0=4.971 atm phi0=1.766 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
1 ANN10	**soft10k**	0.0771	0.146	1.03	59.1	0.273	4.2	0.115	50.7	3.6e+03	0	Complete
1 ANN10	**long**	0.0671	0.119	2.48	1.42	0.405	1.29	0.0318	7.5	279	0	Complete
2 P-only DeepONet10	**base10k**	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	**soft10k**	0.0425	0.0503	0.178	36.4	0.327	1.09	0.0324	3.05	3.37	0	Complete
2 P-only DeepONet10	**long**	5.90e-03	0.0231	0.126	0.551	0.419	0.149	0.0139	0.964	0.896	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.022	0.0336	0.264	12.1	0.32	1.41	0.0334	1.24	6.16	0	Complete
3 ANN10 + P-only DeepONet10	**long**	6.40e-03	0.0238	0.175	0.772	0.423	0.199	0.0133	0.986	1.04	0	Complete
4 T/P DeepONet10	**base10k**	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
5 ANN10 + T/P DeepONet10	**base10k**	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.0574	0.0276	1.07	1.38	0.274	1.42	0.0359	0.966	1.63	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.191	0.0436	3.08	5.53	0.432	5.36	0.132	1.34	4.36	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
6 T/P/ϕ DeepONet10	**long**	0.476	0.561	0.527	144	3.22	11.8	0.57	1.34	2.32	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.468	0.31	2.06	26.2	0.365	22.8	0.562	1.77	2.16	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.314	0.493	0.932	63.3	4.08	10.4	0.47	1.05	1.91	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

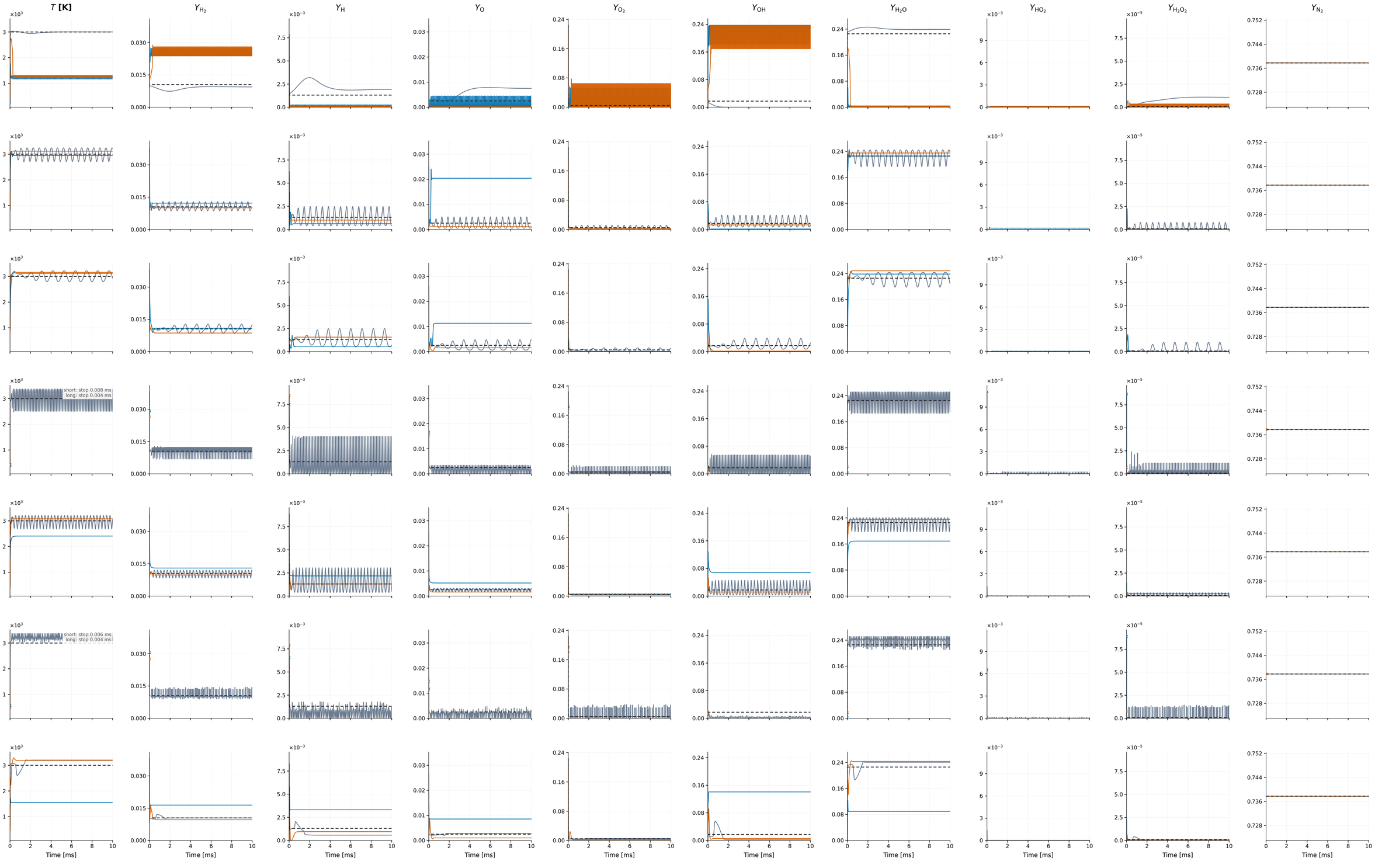
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 76 | Cantera vs NN NRMSE

T0=1523.6 K P0=3.626 atm phi0=1.349 | Repaired self-fed,10ms

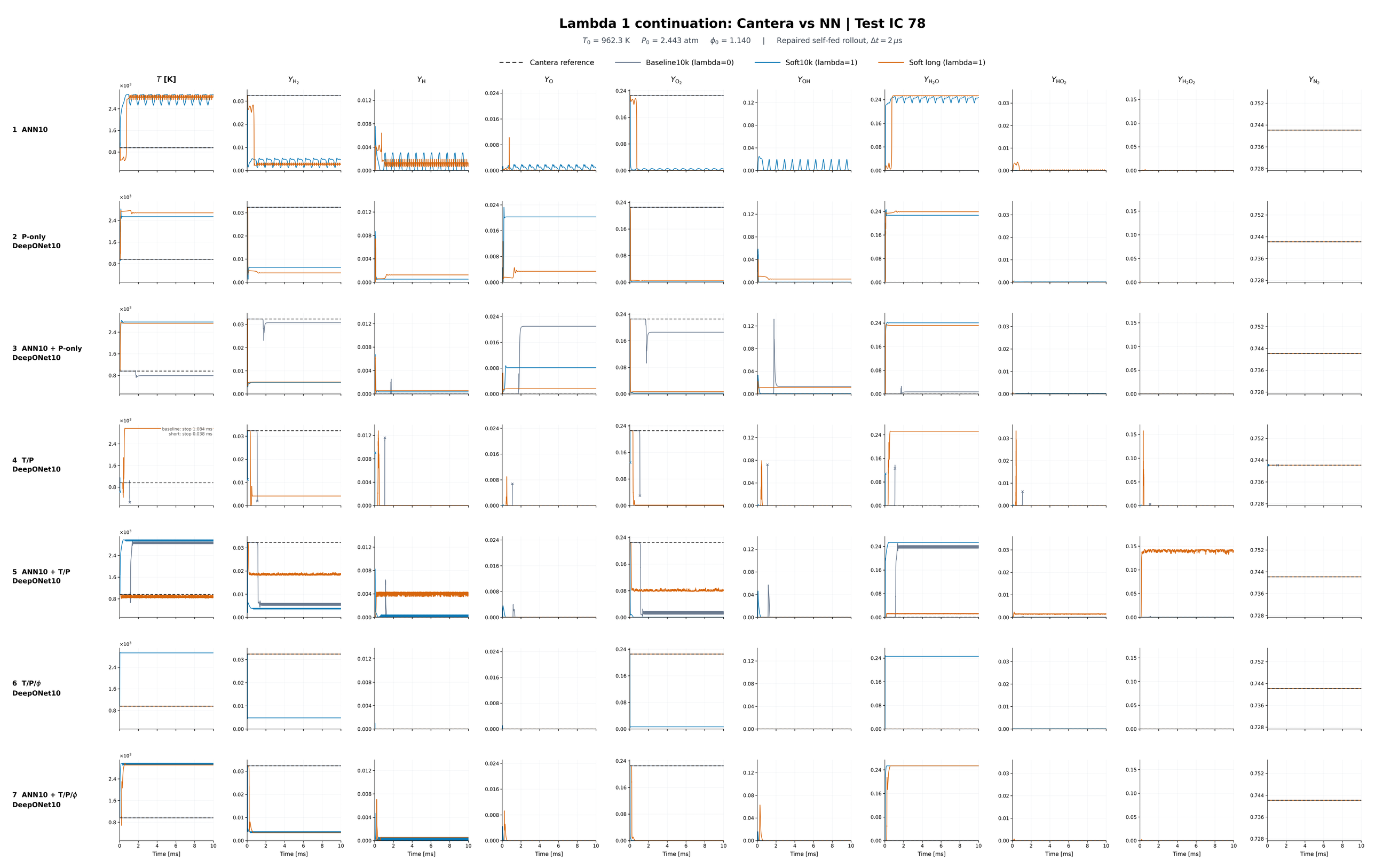
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
1 ANN10	soft10k	0.589	1.35	0.965	0.944	2.61	11.8	0.995	2.84	0.885	0	Complete
1 ANN10	long	0.575	1.38	0.986	0.995	4.02	11.3	0.985	6.25	1.43	0	Complete
2 P-only DeepONet10	base10k	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	soft10k	0.0305	0.182	0.529	7.03	0.971	0.944	0.0413	24.9	1.56	0	Complete
2 P-only DeepONet10	long	0.0401	0.0678	0.235	0.567	0.719	0.257	0.045	0.662	0.794	0	Complete
3 ANN10 + P-only DeepONet10	base10k	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.0521	0.112	0.553	3.43	0.925	1.09	0.0812	8.18	2.23	0	Complete
3 ANN10 + P-only DeepONet10	long	0.053	0.18	0.196	0.408	0.838	0.892	0.101	3.57	1.05	0	Complete
4 T/P DeepONet10	base10k	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
5 ANN10 + T/P DeepONet10	base10k	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.2	0.251	0.669	1.11	0.773	2.9	0.256	5.12	2.84	0	Complete
5 ANN10 + T/P DeepONet10	long	0.0357	0.0771	0.184	0.407	0.626	0.363	0.0478	0.746	0.735	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.006 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.482	0.565	1.55	2.41	0.774	7.02	0.604	4.92	0.898	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.0694	0.105	0.349	0.638	0.733	0.762	0.0848	0.912	0.945	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 78 | Cantera vs NN NRMSE

T0=962.3 K P0=2.443 atm phi0=1.140 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
1 ANN10	soft10k	1.9	0.873	7.43e+06	3.74e+06	0.981	6.05e+07	6.46e+04	16.7	6.24	0	Complete
1 ANN10	long	1.88	0.881	8.11e+06	1.27e+06	0.962	9.12e+05	6.51e+04	161	63	0	Complete
2 P-only DeepONet10	base10k	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	soft10k	1.63	0.8	3.47e+06	7.04e+07	0.983	1.63e+07	6.03e+04	98.6	2.52	0	Complete
2 P-only DeepONet10	long	1.79	0.869	6.36e+06	1.14e+07	0.975	4.37e+07	6.33e+04	8.01	8.37	0	Complete
3 ANN10 + P-only DeepONet10	base10k	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	1.88	0.843	2.52e+06	2.81e+07	0.985	1.85e+07	6.40e+04	46.1	1.02	0	Complete
3 ANN10 + P-only DeepONet10	long	1.83	0.838	3.06e+06	5.70e+06	0.967	7.96e+07	6.17e+04	4.13	1.36	0	Complete
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.084 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
4 T/P DeepONet10	long	2.03	0.857	5.66e+06	1.34e+06	0.974	4.08e+07	6.57e+04	265	1.76e+03	0	Complete
5 ANN10 + T/P DeepONet10	base10k	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	2.07	0.879	2.17e+06	1.28e+06	0.996	2.49e+07	6.73e+04	2.05	1.81	0	Complete
5 ANN10 + T/P DeepONet10	long	0.0951	0.423	2.10e+07	7.00e+04	0.636	1.65e+05	3.26e+03	310	3.00e+04	0	Complete
6 T/P/ϕ DeepONet10	base10k	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	2.04	0.85	1.02e+05	8.15e+04	0.969	4.23e+04	6.56e+04	26.1	8.75	0	Complete
6 T/P/ϕ DeepONet10	long	4.32e-05	2.44e-05	2.02	0.922	5.09e-05	0.975	0.95	0.917	0.955	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	2.07	0.881	1.66e+06	7.52e+05	0.997	6.82e+06	6.75e+04	1.54	2.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	2	0.883	3.29e+06	1.99e+06	0.988	3.76e+07	6.67e+04	12.6	30.8	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

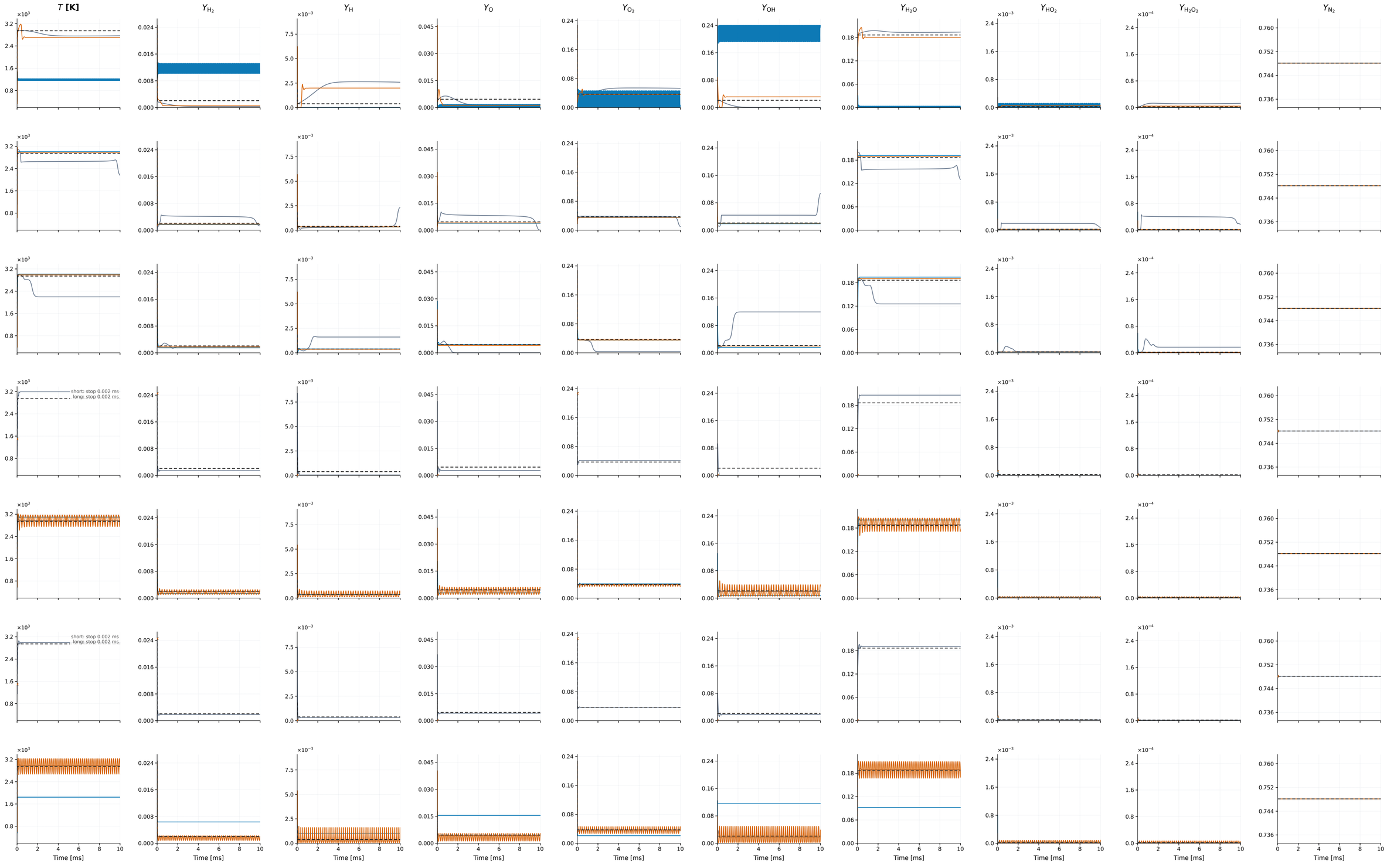
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 88 | Cantera vs NN NRMSE

T0=1502.1 K P0=4.980 atm phi0=0.859 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
1 ANN10	**soft10k**	0.6	4.23	1.01	0.953	0.759	9.78	0.995	1.12	0.969	0	Complete
1 ANN10	**long**	0.0832	0.715	4	0.741	0.07	0.531	0.0409	1.31	1.05	0	Complete
2 P-only DeepONet10	**base10k**	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	**soft10k**	0.0281	0.332	0.17	0.211	0.115	0.124	0.0419	0.529	0.621	0	Complete
2 P-only DeepONet10	**long**	0.0178	0.176	0.248	0.15	0.0798	0.0632	0.0225	0.215	0.239	0	Complete
3 ANN10 + P-only DeepONet10	**base10k**	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.0352	0.411	0.0885	0.121	0.105	0.353	0.0607	0.413	0.797	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0183	0.187	0.187	0.114	0.0741	0.0689	0.0252	0.163	0.257	0	Complete
4 T/P DeepONet10	**base10k**	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
5 ANN10 + T/P DeepONet10	**base10k**	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.0572	0.431	0.175	0.413	0.102	0.683	0.0843	0.592	0.501	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0561	0.309	0.599	0.399	0.0668	0.587	0.0725	0.461	0.648	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
6 T/P/ϕ DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.374	2.04	1.65	2.35	0.435	4.47	0.507	0.627	0.0404	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.071	0.347	1.2	0.444	0.151	0.83	0.0888	1.09	1.27	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.